EV Technology Course — Module 11: Energy Efficiency, Range and Consumption

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Range is not just battery size. It is battery energy divided by how much energy the car uses per kilometre, and that depends on mass, tyres, aerodynamics, drivetrain efficiency, heating and cooling, speed and driving style. This module teaches you to calculate consumption from first principles and to see where every watt-hour goes.

1. Learning objectives

- Calculate road-load forces: rolling resistance, aerodynamic drag, grade and acceleration.
- Convert force and speed into power and Wh/km.
- Trace energy from the grid socket to the wheels and identify losses.
- Explain why EVs are efficient in cities and less so on highways (the opposite of petrol cars).
- Explain test cycles (MIDC, WLTP, EPA) and why real-world range differs.
- Identify the engineering levers that improve efficiency and their cost.

2. Fundamentals

- **Consumption** is measured in Wh/km (or kWh/100 km). A typical Indian compact EV SUV uses about 120–170 Wh/km in real driving.
- **Range = usable battery energy ÷ consumption.** 50 kWh ÷ 150 Wh/km = 333 km.
- Forces the car must overcome: tyre rolling resistance (roughly constant), aerodynamic drag (rises with speed squared), climbing (grade) and acceleration (inertia).
- EVs recover some braking energy through **regeneration**, so stop-start city driving costs less than in a petrol car.
- Heating, cooling and electronics draw power all the time, so their share of energy is largest at low speed.

3. Detailed technical explanation

Road-load forces

Force	Formula	Depends on	Dominant at
Rolling resistance	$m \cdot g \cdot Crr$	Mass, tyre design, pressure, road surface	Low speed
Aerodynamic drag	$\frac{1}{2} \cdot \rho \cdot Cd \cdot A \cdot v^2$	Shape (Cd), frontal area (A), speed	Above ~70–80 km/h
Grade	$m \cdot g \cdot \sin \theta$	Mass, slope	Hills
Inertia	$m \cdot a$	Mass, acceleration	Stop-start traffic (partly recovered by regen)

Typical values

- Crr: 0.007–0.010 for low-rolling-resistance EV tyres; 0.010–0.013 for standard tyres.
- Cd: 0.20–0.25 for very slippery sedans (e.g. Mercedes EQS, Tesla Model 3); 0.30–0.38 for SUVs.
- Frontal area A: 2.1–2.6 m^2 for cars. The product Cd·A is what matters.

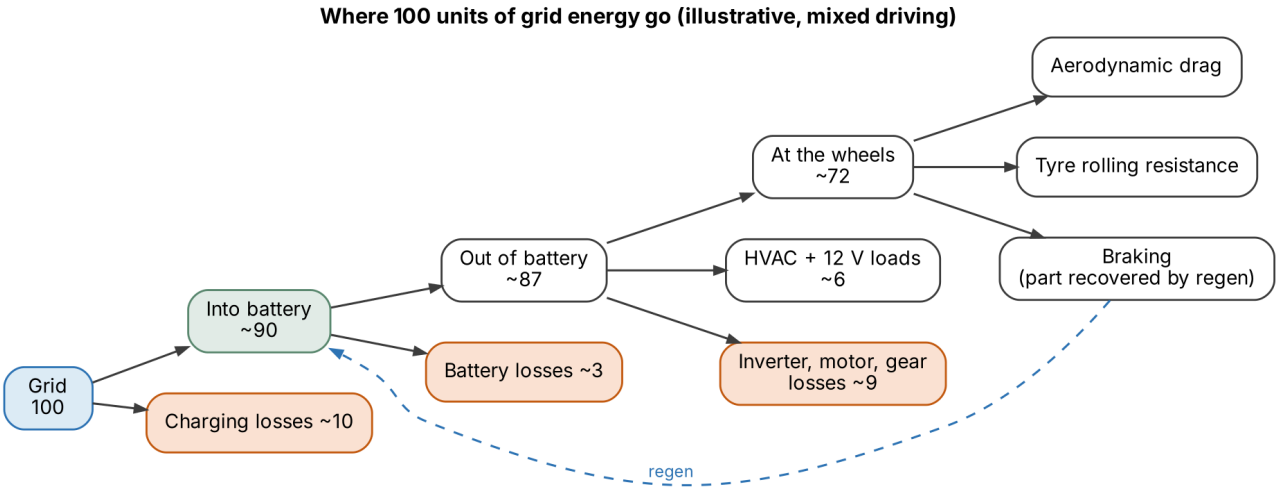
Efficiency chain (grid to wheel)

- Charger and on-board charger losses: ~5–10% (AC charging); DC fast charging loses less in the car but more in the station.
- Battery charge/discharge losses: ~2–5%.
- Inverter: ~97–99%; motor: ~90–96% at good operating points, lower at light loads; gearbox: ~97–98%.
- Drive efficiency (battery to wheel) is typically 85–90% averaged over a cycle.
- Auxiliary loads: HVAC 0.5–5 kW, 12 V systems 0.2–0.5 kW.

Test cycles

- **MIDC** (India) is a gentle, low-speed cycle, so certified ranges are optimistic; real-world range is often 20–35% lower.
- **WLTP** (Europe) is more realistic but still mild; **EPA** (US) includes highway and is closest to real use.
- Real range depends on speed, temperature, AC use, load, tyre pressure and driving style.

4. Diagram: where 100 units of grid energy go



5. Step-by-step: estimating range for a new EV

1. Collect vehicle data: mass with driver, Crr, Cd, A, drive efficiency, auxiliary loads, usable kWh.
2. Choose a drive cycle (city, highway, mixed) or steady speeds.
3. Compute road-load force at each speed.
4. Power at wheel = force × speed; add acceleration and subtract recoverable regen.
5. Divide by drive efficiency to get battery power; add auxiliary power.
6. Integrate over the cycle (or divide by speed for steady state) to get Wh/km.
7. Range = usable kWh ÷ Wh/km; repeat for hot, cold and highway cases to give a range band.

6. Engineering equations

Tractive force:

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F = m g C_{rr} + \tfrac{1}{2}\rho C_d A v^2 + m g \sin\theta + m a
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Wheel power:

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P_{wheel} = F \cdot v
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Battery consumption at steady speed (v in km/h):

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\text{Wh/km} = \frac{F \,[\text{N}]}{3.6\,\eta_{drive}} + \frac{P_{aux}\,
[\text{W}]}{v\,[\text{km/h}]}
```

Range:

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\text{Range} = \frac{E_{usable}}{\text{Wh/km}}
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7. Worked examples

Vehicle: 1,800 kg, Crr 0.009, Cd·A 0.65 m², air density 1.2 kg/m³, drive efficiency 85%.

Example 1: forces at different speeds.

Speed	Rolling (N)	Aero (N)	Wheel power	Wh/km at wheel	Wh/km from battery
50 km/h	159	75	3.3 kW	65	77
100 km/h	159	301	12.8 kW	128	150
120 km/h	159	433	19.7 kW	165	194

Drag quadruples when speed doubles; at 120 km/h aero is almost three times rolling resistance.

Example 2: range. With 50 kWh usable at 150 Wh/km: **333 km**. At 180 Wh/km (hot day, AC, highway): **278 km**.

Example 3: AC impact. A 3 kW AC load adds 3,000 ÷ 30 = **100 Wh/km** in 30 km/h city traffic, but only **30 Wh/km** at 100 km/h. This is why AC hurts city range most.

Example 4: hill. Climbing a 5% grade at 60 km/h needs about 1,800 × 9.81 × 0.05 × 16.7 ≈ **14.7 kW** extra, much of which can be recovered going down.

8. Real-world examples

- **Hyundai Ioniq 6** and **Mercedes EQS** use Cd around 0.21–0.20 to achieve high highway range.
- **Tata Nexon EV, Mahindra BE 6 / XEV 9e, MG Windsor**: Indian test-cycle (MIDC) ranges are well above real-world ranges reported by owners; expect a 20–35% gap.
- **Heat pumps** (Module 7) noticeably improve winter range in cold markets; in India, efficient AC and cabin pre-cooling matter more.
- Low-rolling-resistance tyres and aero wheel covers are common EV efficiency tricks.

9. Manufacturing applications

- **Cost per Wh/km:** every 1% efficiency gain lets the OEM fit a slightly smaller battery for the same range. At battery pack cost of roughly ₹8,000–12,000 per kWh, saving 2 kWh is worth ₹16,000–24,000 per car.
- **Mass reduction** (aluminium, optimised castings) helps rolling resistance and acceleration, but costs money; compare ₹ per kg saved with ₹ per kWh saved.
- **Aero features:** active grille shutters, flush door handles, underbody panels, aero wheels: each has a cost, mass and tooling impact.
- **Tyre sourcing:** low-rolling-resistance EV tyres are more expensive and wear differently.
- **Tolerance and quality:** brake drag, wheel alignment and bearing friction on the line affect real consumption.

10. Common mistakes and misconceptions

- **"A bigger battery always means more range."** Extra mass raises consumption; returns diminish.
- **"Regen recovers all braking energy."** Typically 50–70% of kinetic energy comes back to the battery.
- **"Certified range = real range."** Test cycles are standardised, not representative of every driver.
- **"Drag is minor."** On highways it is the largest single load.

11. Practical case study: the range gap complaint

Problem. Owners of a new compact EV in Delhi report 230 km real range versus 380 km certified (MIDC).

Analysis. Summer AC load of 2–3 kW in traffic adds 60–100 Wh/km; highway trips at 100–110 km/h raise consumption by 40%; heavy loads and under-inflated tyres add more.

Response. Better cabin pre-cooling via app, an efficient AC calibration via OTA, an "eco" AC mode, clearer range estimation in the dashboard and customer education on realistic range.

Lesson. Engineers must design for realistic duty cycles, and marketing must communicate realistic ranges.

12. Revision notes

- Range = usable kWh ÷ Wh/km.
- F = rolling + aero + grade + inertia.
- Aero drag ∝ v^2; power ∝ v^3.
- Drive efficiency 85–90%; regen recovers 50–70% of braking energy.
- AC hurts city range most; speed hurts highway range most.

13. Glossary

Term	Meaning
Crr	Rolling-resistance coefficient
Cd	Drag coefficient
A	Frontal area
Wh/km	Energy consumption per kilometre
MIDC / WLTP / EPA	Test cycles used for certification
Auxiliary load	Power for HVAC, electronics and 12 V systems
Usable capacity	Battery energy the BMS lets you use

14. Knowledge check

Conceptual

1. Why do EVs do relatively well in city driving?
2. Which force dominates at highway speed?
3. Why do test-cycle ranges differ from real ranges?
4. Why does AC affect range more in traffic?
5. Why doesn't a bigger battery give proportionally more range?

Numerical

1. 40 kWh usable, 140 Wh/km. Range?
2. An 1,800 kg car at 80 km/h brakes to a stop. Kinetic energy in Wh? If 60% is recovered, how much?
3. Cd drops from 0.30 to 0.27 on a car with frontal area 2.3 m^2. Power saved at 100 km/h (ρ = 1.2)?

Application

1. As a costing engineer, compare spending ₹3,000 on aero parts versus adding 1 kWh of battery.
2. Propose three ways to close the gap between certified and real range for an Indian EV.

15. Answer key

Conceptual

1. Low speeds mean low drag, regen recovers braking energy, and no idling losses.
2. Aerodynamic drag.
3. Test cycles are gentle and standardised; real driving includes higher speeds, AC, load, temperature and driver style.
4. AC power is spread over fewer kilometres per hour, so Wh/km rises.
5. Added mass increases rolling and inertial losses and cost.

Numerical

1. 40,000 ÷ 140 = **286 km**.
2. ½ × 1,800 × 22.2^2 = 444 kJ = **123 Wh**; 60% recovered = **74 Wh**.
3. Δ(Cd·A) = 0.069 m^2. Power saved = ½ × 1.2 × 0.069 × 27.8^3 ≈ **0.89 kW** (about 9 Wh/km at the wheel).

Application

1. If the aero parts save ~9 Wh/km at highway speed (~5–6%), the same range would otherwise need ~2–3 kWh more battery on a 50 kWh car, costing ₹16,000–36,000. Aero is cheaper, but check tooling, mass and styling.
2. Efficient AC with heat-pump or variable-speed compressor; low-rolling-resistance tyres and aero; better range display and driver coaching; OTA thermal calibrations; publishing realistic ranges.

16. Further learning

- Gillespie, *Fundamentals of Vehicle Dynamics* (road load).
- Ehsani et al., *Modern Electric, Hybrid Electric and Fuel Cell Vehicles*.
- ARAI MIDC and UNECE WLTP regulation texts.
- EPA fueleconomy.gov data for comparing EV consumption.

Next: Module 12, testing, validation and homologation.